

Session 5
30th June 2026

MODEL

VIEW -> input print | Console Based

CONTROLLER -> Algorithmic Implementations -> LOGIC

Operators

Iterations

Conditional Constructs

Session 6
1st July 2026

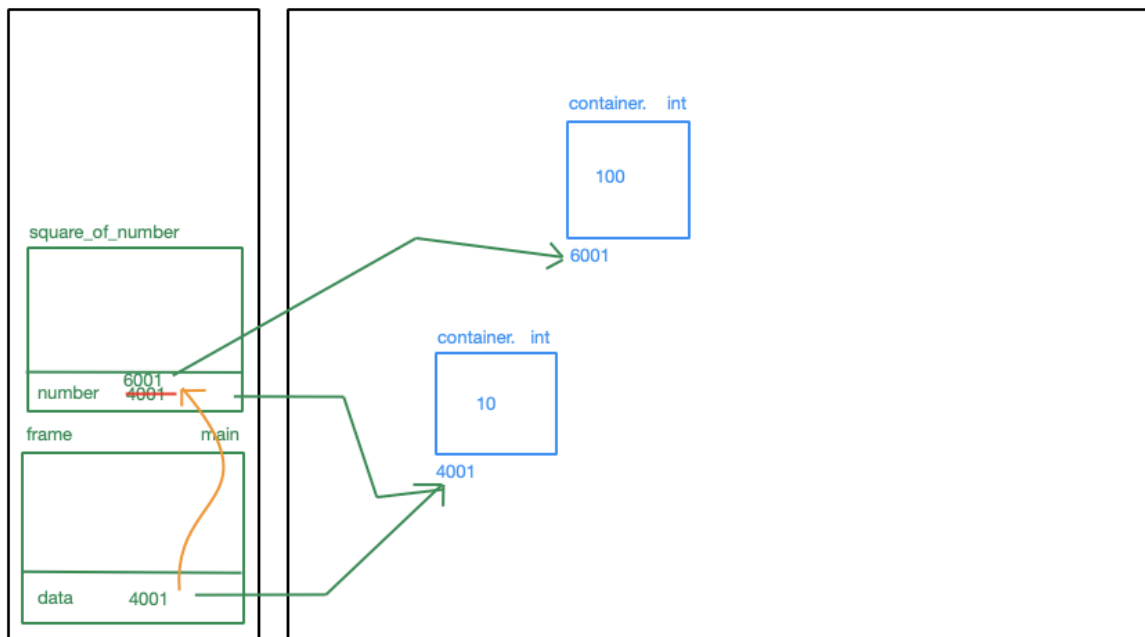
Debugging

Functions

Functions in Memory

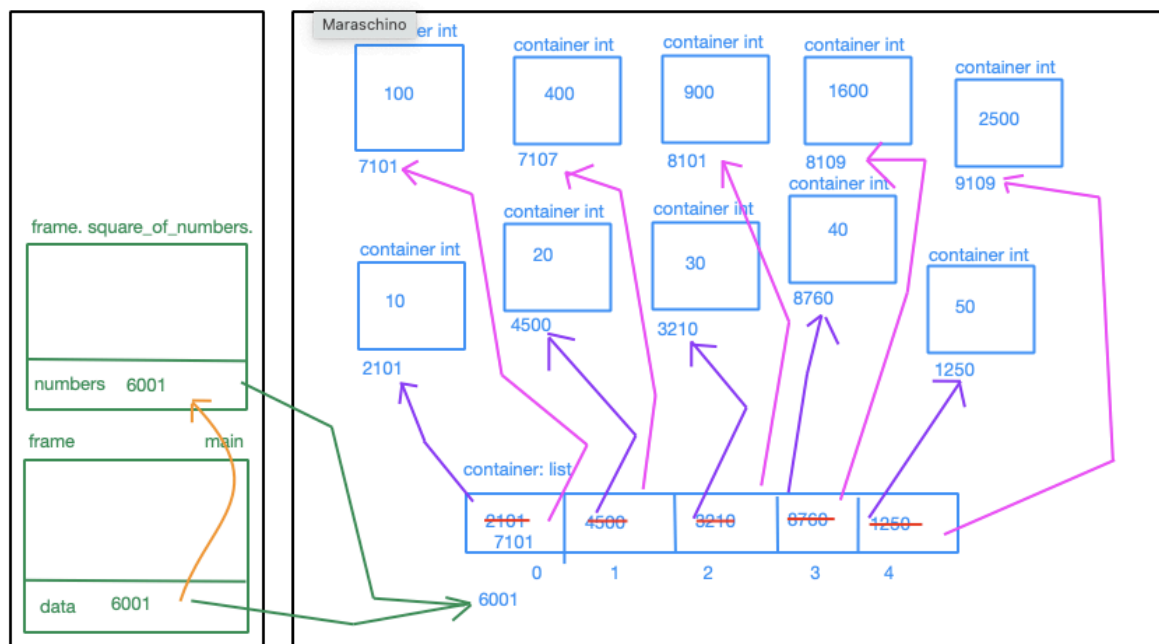
STACK

HEAP



STACK

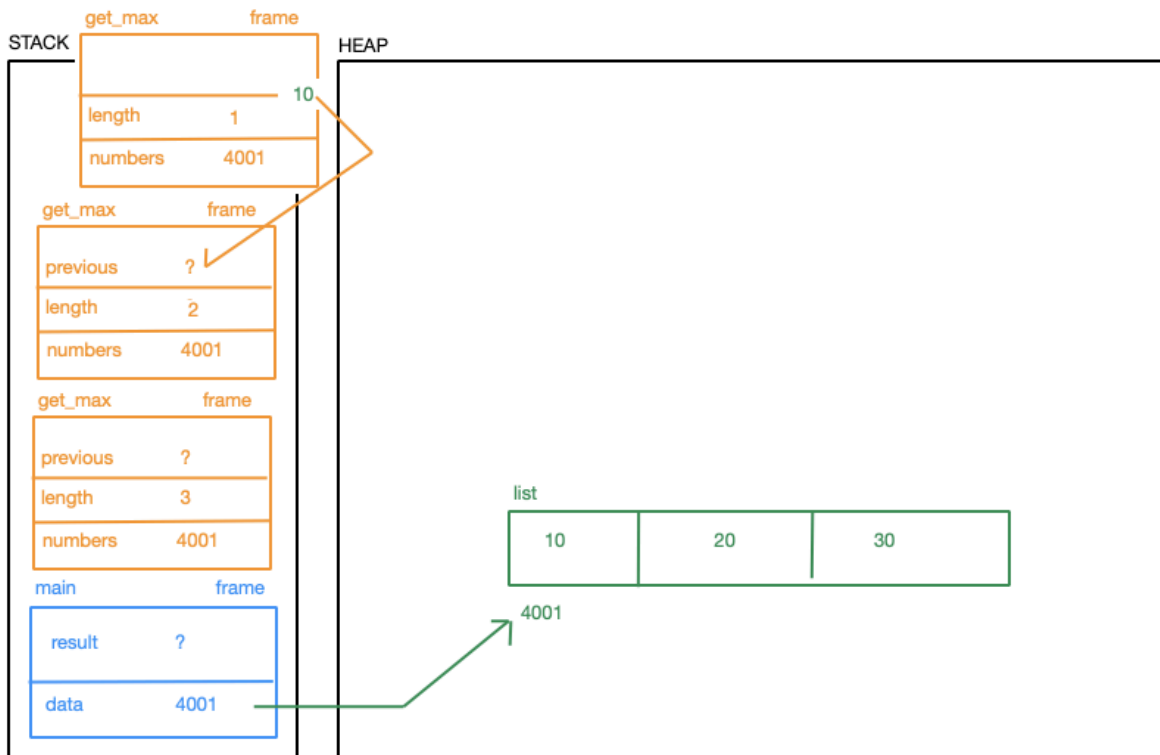
HEAP



Session 7

3rd July 2026

Recursion



Algorithmic Session:
Search/Sort/Filter

Session 8

4th July 2026

OOPS - Object Oriented Programming Structure

2 concepts:

Object

Class

Principle of OOPS:

1. Think of an Object
2. Draw/Represent/Textual Represent the Object
3. From the drawing/representation create a Real Object

Real World

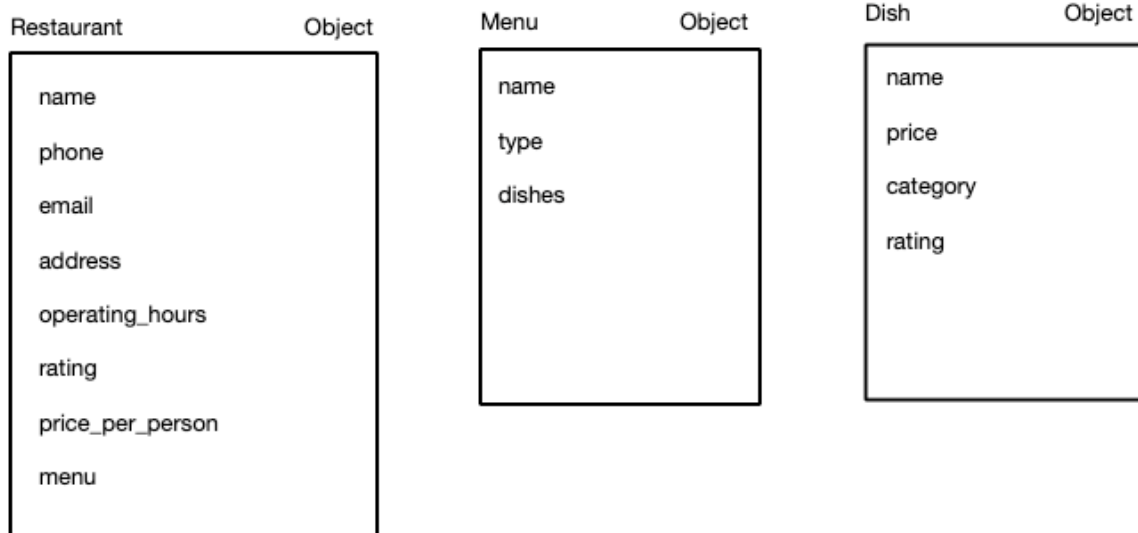
Object -> Anything in existence eg: mango, AC, fan, human

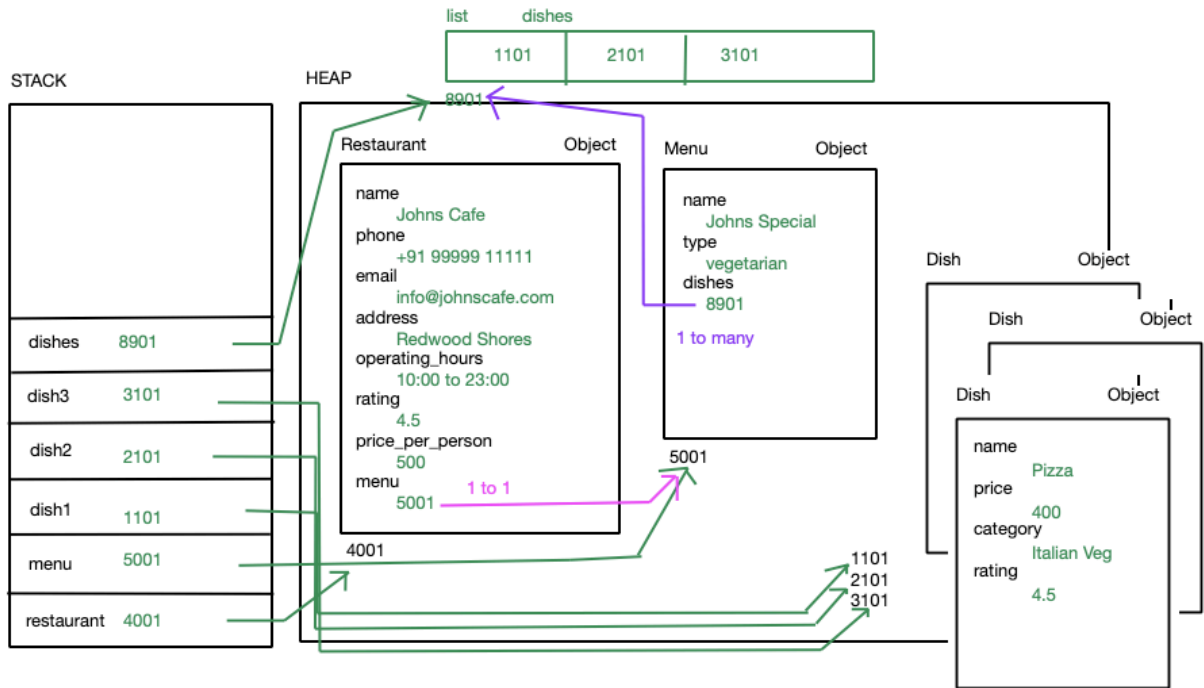
Class -> Drawing/Representation of Object (Architectural Drawing/Blueprint)
Textual Representation

Computer Science World

Object -> Storage Container (Homogeneous/Hetrogeneous)

Class -> Textual Representation of Object, how it will be in memory i.e. the container in memory

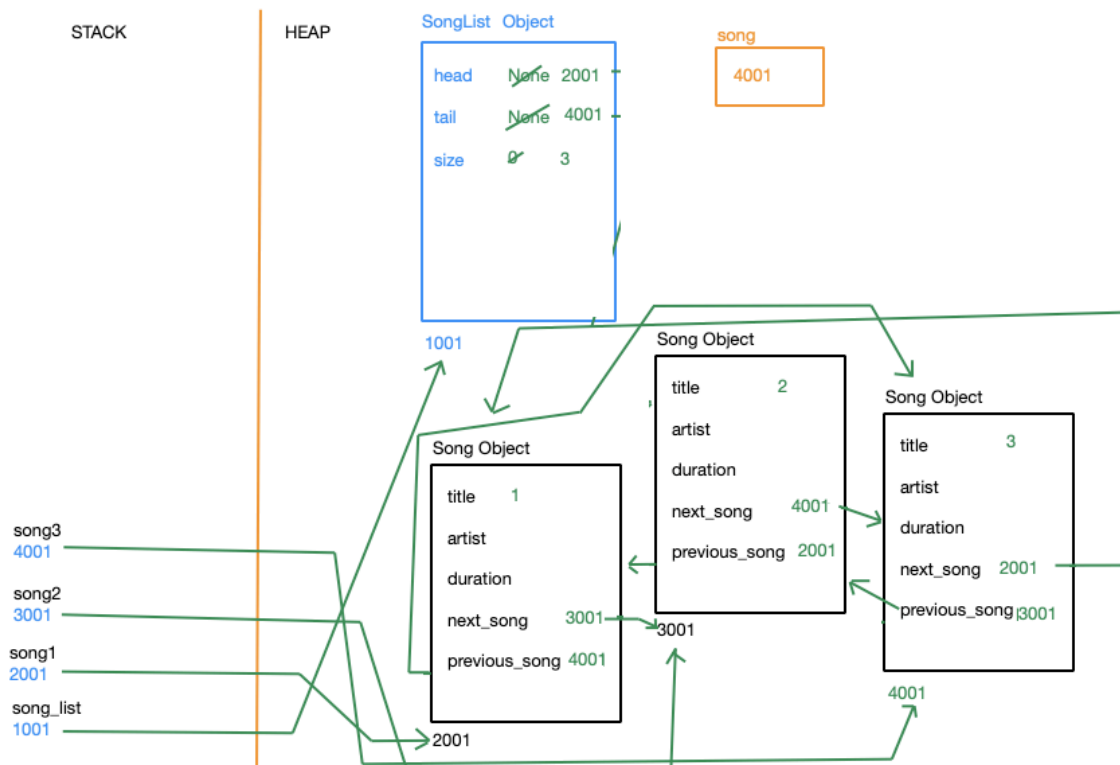




Session 9

6th July 2026

Functions in Class
Coding Circular Doubly LinkedList using OOPS



Session 10

7th July 2026

BackLog:

Stack and Queue DS

Strings in Python

built in functions for list, set, dictionary

lambdas - map filter reduce

iterator and generator

built in packages in python

DataBase Connectivity

Streamlit frontend - chat view database connectivity

Agentic AI